Loan repayments

1 $78 585

2 $1 490 400

3 $600

4 $775

5

a $602 **b** $498.12

6

a $84 039.96 **b** $79 751.24

7

a

i $43 149.33 **ii** $43 299.91 **iii** $43 451.74

b Increasing

c No, because the amount owing is increasing so the instalments are not big enough to pay it off.

8

a $1080

b $54 216

c No, because the balance of the loan will increase instead of decrease.

d Any amount greater than $1080 is acceptable.

9

a No **b** No **c** Yes

10

a $114 444 **b** $113 436 **c** Option 2

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Reducing balance loan tables

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Month	Opening Balance	Interest	Repayment	Closing Balance
1	24 000	162	450	23 712
2	23 712	160.06	450	23 422.06
3	23 422.06	158.1	450	23 130.16

12

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	\$6000	\$480	\$720	\$5760
2	\$5760	\$460.80	\$720	\$5500.80
3	\$5500.80	\$440.06	\$720	\$5220.86

13

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	\$54 000	\$243	\$1500	\$52 743
2	\$52 743	\$237.34	\$1500	\$51 480.34
3	\$51 480.34	\$231.66	\$1500	\$50 212

14

a

Year	Balance owing at beginning of year	Interest charged in this period	Balance owing at end of year	Balance owing after repayment
1	6 500	650	7 150	4 650
2	4 650	465	5 115	2 615
3	2 615	261.5	2 876.5	376.5

b \$465

15

a \$153 971.98 **b** \$1539.72 **c** \$151 521.70

16

a $R = \$900$ **b** 3% **c** \$15 728.70

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17

a 1%

b 4%

c

Quarter	Opening Balance	Interest	Repayment	Closing Balance
1	40 000	400	900	39 500
2	39 500	395	900	38 995
3	38 995	389.95	900	38 484.95

d $\$2700$

18

a 0.35% **b** 4.20% **c** $x = \$420\,000$ **d** $\$1455.77$

19 $\$9517.33$

20

a

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	$\$4500$	$\$180$	$\$2000$	$\$2680$
2	$\$2680$	$\$107.20$	$\$2000$	$\$787.20$
3	$\$787.20$	$\$31.49$	$\$818.69$	$\$0$

b This is the final payment required to pay off the loan. The balance is less than the full payment.

c $\$4818.69$

21

a

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	10 000	1100	2600	8500
2	8500	935	2600	6835
3	6835	751.85	7586.85	0

b $\$2786.85$

c $\$12\,786.85$

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a

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	\$13 600	\$164.02	\$4080	\$9684.02
2	\$9684.02	\$116.79	\$4080	\$5720.81
3	\$5720.81	\$68.99	\$4080	\$1709.80
4	\$1709.80	\$20.62	\$1730.42	\$0

b \$13 970.42

c

Time period (n)	Value at beginning of time period	Interest at end of time period	Repayment this period	Amount at end of time period
1	\$13 600	\$330.01	\$8160	\$5770.01
2	\$5770.01	\$140.01	\$5910.02	\$0

d \$14 070.02

e Half-yearly payments

23

a

	Balance at beginning of year	Interest earned
First year	\$3700	\$259
Second year	\$3959	\$277.13
Third year	\$4236.13	\$296.53
Fourth year	\$4532.06	–

b \$832.66

24

a

Month	Opening Balance (\$)	Interest (\$)	Repayment (\$)	Closing Balance (\$)
1	520 000	2 236	3 750	518 486
2	518 486	2 229.49	3 750	516 965.49
3	516 965.49	2 222.95	3 750	515 438.44
4	515 438.44	2 216.39	3 750	513 904.83
5	513 904.83	2 209.79	3 750	512 364.62

b The balance would be less because they will repay an extra \$3 750 each year.

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25

a

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	2000	2080	750	1330
2	1330	1383.20	750	633.20

b

Time period	Balance owing at beginning of period	Balance owing plus interest during period	Repayment	Balance owing at end of period
1	2000	2040	375	1665
2	1665	1698.30	375	1323.30
3	1323.30	1349.77	375	974.77
4	974.77	994.27	375	619.27

c $\$13.93$

26

a 1.52%

b

i $X = 3173.67$ **ii** $Y = 47.83$ **iii** $Z = 0.39$ **iv** $W = 25.86$

v $\$2000.86$

27

a 0.5%

b

Quarter	Opening Balance	Interest	Repayment	Closing Balance
72	127.81	0.64	128.45	0

c 6 years

d $\$17\,878.45$

e $\$2878.45$

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28

a 0.75%

b

Quarter	Opening Balance	Interest	Repayment	Closing Balance
151	2081.99	15.61	2097.60	0

c 12.58 years

d $\$752\,097.60$

e $\$450\,000$

Financial tables

29 $\$383.54$

30 $\$401.28$

31 $\$7956$

32

a $\$391\,170$ **b** $\$379\,134$ **c** Option 2

33 Option 1, because she will save $\$10\,745$ in repayments and $\$9305$ in fees.

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